

Podcast 06 — VPC Connectivity Patterns

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Length target: 15 min (~2,200 spoken words) **Repo references:** `domain-1-organizational/08-vpc.md` (TGW + PrivateLink + Endpoints), `10-aws-privatelink.md`, `32-vpc-peering.md`

Topic & Scope

Connecting VPCs together and reaching AWS services privately. The decision matrix between VPC Peering, Transit Gateway, PrivateLink, and VPC Endpoints is the highest-yield SAP-C02 networking knowledge.

Service Coverage Depth

Deep: - VPC Peering — 1:1, non-transitive, simple - Transit Gateway — hub-and-spoke, route tables, attachments - PrivateLink (VPC Interface Endpoints) — one-way service exposure - VPC Endpoints — Gateway (S3, DynamoDB) vs Interface (everything else)

Brief: - VPC Lattice (mention as the “service-to-service mesh across VPCs” newer option)

Structured Outline

1. **Open (30s)** — “Four ways to connect VPCs. Picking the wrong one = wasted money or broken architecture.”
2. **VPC Peering (3 min)** — 1:1 connection, manual routing, no transitive, same/cross-Region/cross-account
3. **Transit Gateway (4 min)** — hub-and-spoke, scales to thousands of VPCs, TGW route tables, attachments (VPC, VPN, DX, peering), inter-region peering
4. **PrivateLink (3 min)** — expose service in your VPC via NLB to other VPCs via Interface Endpoint, one-way, no peering / no overlapping CIDR issues
5. **VPC Endpoints (3 min)** — Gateway (S3, DynamoDB — free, route table entry) vs Interface (any AWS service — ENI + DNS, hourly cost), endpoint policies
6. **Decision Matrix (1.5 min)**
7. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- VPC Peering: 1:1, non-transitive, same or cross-Region or cross-account
- Peering requires non-overlapping CIDRs and explicit route table entries
- Transit Gateway: hub-and-spoke for many VPCs, transitive routing through TGW route tables
- TGW supports VPC, VPN, Direct Connect Gateway, and TGW peering attachments
- TGW inter-region peering: encrypted over AWS backbone

- PrivateLink exposes a service from one VPC to consumer VPCs via NLB + Interface Endpoint
- PrivateLink does NOT require non-overlapping CIDRs (uses DNS name + private IPs in consumer VPC)
- Gateway VPC Endpoint: S3 + DynamoDB only, FREE, adds prefix list to route table
- Interface VPC Endpoint: ENI in your subnet, supports most AWS services + Marketplace, hourly + data processing charges
- Endpoint policies further restrict what the endpoint allows
- Resource shares of TGW + Resolver rules via RAM
- VPC Lattice: application-layer service-to-service mesh — newer option for HTTP/gRPC across VPCs/accounts

Must-Mention Exam Traps

- EXAM TRAP: VPC Peering is NOT transitive — $A \leftrightarrow B + B \leftrightarrow C$ does NOT give $A \leftrightarrow C$
- EXAM TRAP: VPC Peering with overlapping CIDRs is impossible
- EXAM TRAP: Transit Gateway route tables are DIFFERENT from VPC route tables — both must be configured
- EXAM TRAP: TGW attachment costs are hourly + data — at scale TGW gets expensive vs PrivateLink for one-service exposure
- EXAM TRAP: PrivateLink is ONE-WAY — provider exposes, consumer connects; reverse needs another endpoint
- EXAM TRAP: Gateway Endpoint is FREE; Interface Endpoint costs hourly per AZ + per-GB
- EXAM TRAP: Gateway Endpoint only works for S3 and DynamoDB — everything else is Interface Endpoint
- EXAM TRAP: VPC Endpoint policy + IAM identity policy + bucket policy ALL apply — most restrictive wins
- EXAM TRAP: Direct Connect Gateway allows DX to reach multiple VPCs across Regions — but TGW is the modern path
- EXAM TRAP: PrivateLink doesn't traverse the public internet — but you still pay for endpoint hours + data
- EXAM TRAP: For S3 from many VPCs across Regions, use S3 Multi-Region Access Points + Gateway Endpoints — not PrivateLink

Key Decision Matrix

Scenario	Pick
Connect 2 VPCs, same workload, simple	VPC Peering
Connect 50+ VPCs in mesh	Transit Gateway
Expose one service to 100 consumer VPCs	PrivateLink (Interface Endpoint)
Access S3 / DynamoDB from private subnet without NAT	Gateway VPC Endpoint

Scenario	Pick
Access any AWS API privately from VPC	Interface VPC Endpoint
Overlapping CIDRs need to talk	PrivateLink (peering won't work)
Hybrid + multi-VPC + multi-account routing	Transit Gateway

Tone & Style

- Frame as four-way decision tree
- Use the “non-transitive” property of peering as the recurring trap
- For Gateway vs Interface Endpoint: “S3 + DynamoDB are special — they get free Gateway Endpoints. Everything else is Interface.”

Rapid-Fire Closer

“Peering transitive?” — “NO.” “Free endpoint for S3?” — “Gateway Endpoint.” “Expose one service to many VPCs?” — “PrivateLink.” “50 VPCs to connect?” — “Transit Gateway.” “Overlapping CIDRs?” — “PrivateLink, not peering.”